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Class:SY B

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EXPERIMENT NO 1.5:

Scanner Class, Getter Setter Methods.

Scanner Class

1. Write a program that takes 5 integers as input from the user using the Scanner class, stores them in an array, and then prints all the elements.

```
import java.util.Scanner;


public class PrintFiveNumbers {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);
      int arr[] = new int[5];


      System.out.println("Enter 5 integers:");


      for(int i = 0; i < 5; i++) {
         arr[i] = sc.nextInt();
      }


      System.out.println("The elements are:");
      for(int i = 0; i < 5; i++) {
         System.out.print(arr[i] + " ");
      }
```

```
    sc.close();
  }
}
```

```
Microsoft Windows [Version 10.0.26200.7840]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Swati\OneDrive\Desktop\java>javac PrintFiveNumbers.java

C:\Users\Swati\OneDrive\Desktop\java>java PrintFiveNumbers
Enter 5 integers:
1
2
3
4
5
The elements are:
1 2 3 4 5
C:\Users\Swati\OneDrive\Desktop\java>
```

2. Read n integers into an array. Compute and display the sum and average of the elements.

```
import java.util.Scanner;


public class SumAverage {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);


    System.out.print("Enter number of elements: ");
    int n = sc.nextInt();


    int arr[] = new int[n];
    int sum = 0;
```

```
    System.out.println("Enter elements:");
    for(int i = 0; i < n; i++) {
      arr[i] = sc.nextInt();
      sum = sum + arr[i];
    }

    double average = (double) sum / n;

    System.out.println("Sum = " + sum);
    System.out.println("Average = " + average);

    sc.close();
  }
}
```

```
C:\Users\Swati\OneDrive\Desktop\java>java SumAverage
Enter number of elements: 5
Enter elements:
2
6
1
9
2
Sum = 20
Average = 4.0

C:\Users\Swati\OneDrive\Desktop\java>
```

3. Read n integers into an array. Count and display how many numbers are even and how many are odd.

```
import java.util.Scanner;
```

```
public class EvenOddCount {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    System.out.print("Enter number of elements: ");
    int n = sc.nextInt();

    int arr[] = new int[n];
    int even = 0, odd = 0;

    System.out.println("Enter elements:");
    for(int i = 0; i < n; i++) {
      arr[i] = sc.nextInt();

      if(arr[i] % 2 == 0)
        even++;
      else
        odd++;
    }

    System.out.println("Even count = " + even);
    System.out.println("Odd count = " + odd);

    sc.close();
  }
}
```

```
C:\Users\Swati\OneDrive\Desktop\java>java EvenOddCount
Enter number of elements: 5
Enter elements:
23
46
12
23
12
Even count = 3
Odd count = 2
```

4. Input an array of integers. Sort the array in ascending order and display the sorted result.

```
import java.util.Scanner;


public class SortArray {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);


    System.out.print("Enter number of elements: ");
    int n = sc.nextInt();


    int arr[] = new int[n];


    System.out.println("Enter elements:");
    for(int i = 0; i < n; i++) {
      arr[i] = sc.nextInt();
    }


    for(int i = 0; i < n - 1; i++) {
      for(int j = 0; j < n - i - 1; j++) {
        if(arr[j] > arr[j + 1]) {
```

```
            int temp = arr[j];
            arr[j] = arr[j + 1];
            arr[j + 1] = temp;
        }
    }
  }

  System.out.println("Sorted array:");
  for(int i = 0; i < n; i++) {
    System.out.print(arr[i] + " ");
  }

  sc.close();
 }
}
```

```
C:\Users\Swati\OneDrive\Desktop\java>java SortArray
Enter number of elements: 5
Enter elements:
12
34
10
5
23
Sorted array:
5 10 12 23 34
```

5. Accept two arrays of integers from the user. Merge them into a single array and display the merged array.

```
import java.util.Scanner;
```

```
public class MergeArrays {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    System.out.print("Enter size of first array: ");
    int n1 = sc.nextInt();

    int arr1[] = new int[n1];
    System.out.println("Enter elements of first array:");
    for(int i = 0; i < n1; i++) {
      arr1[i] = sc.nextInt();
    }

    System.out.print("Enter size of second array: ");
    int n2 = sc.nextInt();

    int arr2[] = new int[n2];
    System.out.println("Enter elements of second array:");
    for(int i = 0; i < n2; i++) {
      arr2[i] = sc.nextInt();
    }

    int merged[] = new int[n1 + n2];

    // Copy first array
    for(int i = 0; i < n1; i++) {
      merged[i] = arr1[i];
    }
```

```
        // Copy second array
        for(int i = 0; i < n2; i++) {
            merged[n1 + i] = arr2[i];
        }

        System.out.println("Merged array:");
        for(int i = 0; i < merged.length; i++) {
            System.out.print(merged[i] + " ");
        }

        sc.close();
    }
}
```

```
C:\Users\Swati\OneDrive\Desktop\java>java MergeArrays
Enter size of first array: 5
Enter elements of first array:
2
4
1
6
7
Enter size of second array: 3
Enter elements of second array:
7
2
3
Merged array:
2 4 1 6 7 7 2 3
```

Getter Setter Methods.

1. Student Information

Create a Student class with private fields:

name and rollNumber.

Write getter and setter methods for both fields.

In the main method, accept student details using Scanner, set them using setters, and display them using getters.

```
import java.util.Scanner;


class Student {
   private String name;
   private int rollNumber;


   // Setter methods
   public void setName(String name) {
      this.name = name;
   }


   public void setRollNumber(int rollNumber) {
      this.rollNumber = rollNumber;
   }


   // Getter methods
   public String getName() {
      return name;
   }


   public int getRollNumber() {
      return rollNumber;
   }
}
```

```
public class StudentMain {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      Student s = new Student();

      System.out.print("Enter Name: ");
      s.setName(sc.nextLine());

      System.out.print("Enter Roll Number: ");
      s.setRollNumber(sc.nextInt());

      System.out.println("\nStudent Details:");
      System.out.println("Name: " + s.getName());
      System.out.println("Roll Number: " + s.getRollNumber());

      sc.close();
   }
}
```

```
C:\Users\Swati\OneDrive\Desktop\java>java StudentMain
Enter Name: Swati
Enter Roll Number: 27

Student Details:
Name: Swati
Roll Number: 27
```

2. Bank Account Balance

Define a BankAccount class with private fields: accountNumber and balance.

Provide setters to initialize values and getters to retrieve them.

Add a method deposit (double amount) and withdraw (double amount) that internally update the balance using setters.

Demonstrate usage in the main method.

```
import java.util.Scanner;


class BankAccount {
   private String accountNumber;
   private double balance;


   public void setAccountNumber(String accountNumber) {
      this.accountNumber = accountNumber;
   }


   public void setBalance(double balance) {
      this.balance = balance;
   }


   public String getAccountNumber() {
      return accountNumber;
   }


   public double getBalance() {
      return balance;
   }


   public void deposit(double amount) {
      balance = balance + amount;
```

```
    }

    public void withdraw(double amount) {
        if(amount <= balance)
            balance = balance - amount;
        else
            System.out.println("Insufficient Balance!");
    }
}

public class BankMain {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        BankAccount acc = new BankAccount();

        System.out.print("Enter Account Number: ");
        acc.setAccountNumber(sc.nextLine());

        System.out.print("Enter Initial Balance: ");
        acc.setBalance(sc.nextDouble());

        acc.deposit(1000);
        acc.withdraw(500);

        System.out.println("Account Number: " + acc.getAccountNumber());
        System.out.println("Balance: " + acc.getBalance());

        sc.close();
```

```
    }
}
```

```
C:\Users\Swati\OneDrive\Desktop\java>java BankMain
Enter Account Number: 2456413876
Enter Initial Balance: 500
Account Number: 2456413876
Balance: 1000.0
```

3. Employee Salary Update

Create an Employee class with private fields: empId, empName, and salary.

Use getters and setters to access/update these fields.

Write a method increaseSalary(double percentage) that modifies salary using setters.

Accept employee details from the user and show the updated salary.

```
import java.util.Scanner;


class Employee {
    private int empId;
    private String empName;
    private double salary;


    public void setEmpId(int empId) {
        this.empId = empId;
    }


    public void setEmpName(String empName) {
        this.empName = empName;
    }


    public void setSalary(double salary) {
```

```
        this.salary = salary;
    }

    public int getEmpId() {
        return empId;
    }

    public String getEmpName() {
        return empName;
    }

    public double getSalary() {
        return salary;
    }

    public void increaseSalary(double percentage) {
        salary = salary + (salary * percentage / 100);
    }
}

public class EmployeeMain {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Employee emp = new Employee();

        System.out.print("Enter Employee ID: ");
        emp.setEmpId(sc.nextInt());
        sc.nextLine();
```

```
        System.out.print("Enter Employee Name: ");
        emp.setEmpName(sc.nextLine());

        System.out.print("Enter Salary: ");
        emp.setSalary(sc.nextDouble());

        System.out.print("Enter Salary Increase Percentage: ");
        double percent = sc.nextDouble();

        emp.increaseSalary(percent);

        System.out.println("\nUpdated Salary: " + emp.getSalary());

        sc.close();
    }
}
```

```
C:\Users\Swati\OneDrive\Desktop\java>java EmployeeMain
Enter Employee ID: 12435
Enter Employee Name: Swati
Enter Salary: 265768
Enter Salary Increase Percentage: 25

Updated Salary: 332210.0
```

4. Product Inventory

Define a Product class with private fields: productId, productName, and quantity.

Implement getters and setters for all fields.

Add a method updateQuantity(int qty) that uses setters to change the quantity.

In the main method, simulate adding/removing stock.

```
import java.util.Scanner;

class Product {
  private int productId;
  private String productName;
  private int quantity;

  public void setProductId(int productId) {
    this.productId = productId;
  }

  public void setProductName(String productName) {
    this.productName = productName;
  }

  public void setQuantity(int quantity) {
    this.quantity = quantity;
  }

  public int getProductId() {
    return productId;
  }

  public String getProductName() {
    return productName;
  }
```

```
    public int getQuantity() {
        return quantity;
    }

    public void updateQuantity(int qty) {
        quantity = quantity + qty;
    }
}

public class ProductMain {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Product p = new Product();

        System.out.print("Enter Product ID: ");
        p.setProductId(sc.nextInt());
        sc.nextLine();

        System.out.print("Enter Product Name: ");
        p.setProductName(sc.nextLine());

        System.out.print("Enter Initial Quantity: ");
        p.setQuantity(sc.nextInt());

        System.out.print("Enter Quantity to Add/Remove: ");
        int qty = sc.nextInt();
```

```
        p.updateQuantity(qty);

        System.out.println("Updated Quantity: " + p.getQuantity());

        sc.close();
    }
}
```

```
C:\Users\Swati\OneDrive\Desktop\java>java ProductMain
Enter Product ID: 2313
Enter Product Name: car
Enter Initial Quantity: 237698
Enter Quantity to Add/Remove: 2
Updated Quantity: 237700
```

5. Car Details with Validation

Create a Car class with private fields: brand, model, and price.

Implement getters and setters, but in the setter for price, ensure that the value cannot be negative (add validation logic).

Accept car details from the user, set them, and display the details using getters.

```
import java.util.Scanner;

class Car {
    private String brand;
    private String model;
    private double price;

    public void setBrand(String brand) {
        this.brand = brand;
    }
```

```
    public void setModel(String model) {
        this.model = model;
    }

    public void setPrice(double price) {
        if(price >= 0)
            this.price = price;
        else
            System.out.println("Price cannot be negative!");
    }

    public String getBrand() {
        return brand;
    }

    public String getModel() {
        return model;
    }

    public double getPrice() {
        return price;
    }
}

public class CarMain {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
```

```
        Car car = new Car();

        System.out.print("Enter Brand: ");
        car.setBrand(sc.nextLine());

        System.out.print("Enter Model: ");
        car.setModel(sc.nextLine());

        System.out.print("Enter Price: ");
        car.setPrice(sc.nextDouble());

        System.out.println("\nCar Details:");
        System.out.println("Brand: " + car.getBrand());
        System.out.println("Model: " + car.getModel());
        System.out.println("Price: " + car.getPrice());

        sc.close();
    }
}
```

```
C:\Users\Swati\OneDrive\Desktop\java>java CarMain
Enter Brand: mercedes
Enter Model: 123
Enter Price: 30000000

Car Details:
Brand: mercedes
Model: 123
Price: 3.0E7
```